Chenyu Gu

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Research Interests

Medical & Bio-Robotics:

- Surgical robots, wearable and rehabilitation robotics, soft robotics, data-driven robotic control.

Medical Devices & Digital Health:

- Wearable/portable devices, surgical navigation, tele-rehabilitation.

Biomechanics & Physiological Signals:

- Gait and locomotion analysis, motion tracking, multi-modal EMG/IMU sensing and analysis.

Education

Washington University in St. Louis — Saint Louis, United States
Ph.D. *in Biomedical Engineering;* — *Aug 2026 – Present*
Advisor: Dr. Ismael Seanez

Georgia Institute of Technology — Atlanta, United States
M.S. *in Bioengineering (ECE);* ***GPA: 4.00/4.00*** — *Aug 2024 – May 2026*
Relevant coursework: Digital Control System, Finite Element Analysis.
Advisor: Dr. Jaydev Desai; Thesis: Design, modeling, and control of a polymer-based continuum robot

Southern University of Science and Technology — Shenzhen, China
B.S. *in Intelligent Medical Engineering;* ***GPA: 3.82/4.00 Major Related: 3.88/4.00*** — *Aug 2020 – Jun 2024*
Relevant coursework: Medical Robotics, Principles of Electrical Circuits, Signals and Systems, Digital Signal Processing, Medical Imaging, Machine Learning, Neural Engineering, BCI.
Advisor: Dr. Mingming Zhang; Thesis: Design and Validation of an Inertial-based Navigation System

KTH Royal Institute of Technology — Stockholm, Sweden
Exchange student in Electrical Engineering and Computer Science — *Sep 2023 – Jan 2024*
Relevant coursework: Principles of Wireless Sensor Networks, Social Robotics.
Advisor: Dr. Seraina Dual

Publications & Patents

- T. A. Brumfiel, R. Konda, **C. Gu**, and J. P. Desai. Design and Modeling of a Modified COAST Guidewire Robot with a Concentric Push-Pull Mechanism. npj Robotics. (**Accepted**)
- **C. Gu**, T. A. Brumfiel, N. Malhotra, and J. P. Desai, "Design and Modeling of a Polymer-based Hydraulic Continuum Robot for Minimal Invasive Surgery," International Symposium on Medical Robotics (ISMR), 2026. (**Best Paper Award**)
- **Chenyu Gu**, Weicong Lin, Xinyi He, Lei Zhang, Mingming Zhang, "IMU-based motion capture system for rehabilitation applications: A systematic review," *Biomimetic Intelligence and Robotics*, vol. 3, no. 2, p. 100097, 2023. DOI: 10.1016/j.birob.2023.100097. (Annual Best Paper Award) **(IF: 5.4)**
- **Chenyu Gu**, Yangyi Yu, Xinyi He, Lei Zhang, Zhen Xi, Yudong Liu, Guangheng Li, and Mingming Zhang, "A Portable Inertial Navigation System for Total Hip Arthroplasty Targeting Direct Anterior Approach," in *IEEE Transactions on Instrumentation and Measurement*, doi: 10.1109/TIM.2025.3548784. **(IF: 5.9)**
- Li Guangheng, Zhang Mingming, Yu Yangyi, and **Chenyu Gu**. A joint replacement simulation device and a joint replacement simulation method. China CN118634071A, filed July 9, 2024, and issued September 13, 2024.
- Zhang Mingming, and **Chenyu Gu**. Human motion capture method, system and storage medium. China CN118870291A, filed July 9, 2024, and issued October 29, 2024.

Research Experience

Spatiotemporal control of spinal cord stimulation Saint Louis, United States

Project Leader in Seanez Lab ***Skills: Human Study & Physiological Signal Processing*** *Aug 2026 – Present*

- Led clinical protocol execution across 16 chronic spinal cord injury participants to evaluate multielectrode transcutaneous spinal cord stimulation arrays, optimizing spatial positioning to increase targeted joint torque.
- Developed frequency-modulated software to evaluate posterior root-muscle reflexes via post-activation depression while monitoring real-time autonomic nociception and pain profiles.
- Integrated a 16-channel wireless EMG system, Biodex isokinetic dynamometer, to synchronize high-density neurophysiological, torque, and 3D kinematic data.
- Implemented Linear Mixed-Effects Models within a 15-session crossover study design to analyze repeated-measures biomechanical outcomes and muscle recruitment selectivity.

Design, modeling, and control of a hydraulically actuated robot Atlanta, United States

Project Leader in RoboMed Lab ***Skills: CAD Modeling, Robotics Modeling & Control*** *Oct 2024 – May 2026*

- Develop and manufacture a polymer-based hydraulic robotic system for non-conductive, MRI safe environment.
- Formulate the physical model for the hydraulic robot and refine it with a data-driven hysteresis model.
- Implement model-based open-loop control and integrate shape sensing methods for close-loop control.
- Conduct experiments to characterize and validate the system, and demonstrate it under a vascular model.

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Mechanism and actuation design of COAST Guidewire Robot Atlanta, United States

Project Assistant in RoboMed Lab ***Skills: Circuit & PCB Design*** *Jul 2025 – March 2026*

- Contribute to the development of a modified COAST robotic guidewire enabling active stiffness control to enhance steering and quick connect usability in minimally invasive cardiovascular interventions. (Paper prepared for IJRR)
- Manage the electrical development of the actuation system, including PCB and circuit design, and develop analog force sensor to enable precise robotic actuation and sensing.

Navigation System for Surgeries Shenzhen, China

Project Leader in Brain-robot Rehabilitation Lab ***Skills: Software & Hardware Design*** *Feb 2023 – Jun 2024*

- Hardware and software co-designed a surgery navigation system, including circuit design to integrate micro-controller and sensor, C embedded programming, and real-time tracking feedback through Unity.
- Conduct experiments to validate effectiveness in both validity and reliability with optical system and robot arm.

Movement Analysis using IMU Sensors Stockholm, Sweden

Visiting Student in Intelligent Heart Technology Lab ***Skills: Kalman Filter & Sensor Fusion*** *Oct 2023 – Jan 2024*

- Investigated relationship between movement and cardiovascular function through gait parameters and heart rate.
- Deployed a Kalman filter for orientation estimation, calibrated IMU angles, and extracted gait parameter and CoM.

Motion Capture for Rehabilitation Shenzhen, China

Project Leader in Brain-robot Rehabilitation Lab ***Skills: Signal Processing, Deep Learning*** *Apr 2022 – May 2024*

- Integrated UWB technology and IMU to develop a motion capture system for human motion recovery.
- Developed Unity visualization and Python signal transmission, and applied deep learning for reconstruction.
- Evaluated the system's effectiveness on healthy subjects during rehabilitation exercises.

Center of Mass Estimation and Rehabilitation Applications Shenzhen, China

Project Assistant in Brain-robot Rehabilitation Lab ***Skills: Data Analysis, Acquisition*** *Oct 2022 – Aug 2023*

- Utilized IMU-based motion capture and developed Opensim-based algorithm to estimate the 3D center of mass.
- Analyzed the center of mass to extract motion features and perform rehabilitation assessments.

Working & Teaching Experience

Graduate Teaching Assistant at Georgia Tech — Atlanta, United States
GTA for Machine Learning (CS 7641) — *Aug 2025 – May 2026*

- Organize weekly office hours and review sessions to clarify concepts in machine learning, and instruct students to help them improve and success in the course.
- Review homework and projects to guide and correct misunderstanding for over 300 graduate students.

Department of Orthopedics, First People's Hospital — Shenzhen, China
Research Assistant — *Jun 2023 – Aug 2023*

- Conducted an investigation to identify suitable medical measurement devices to support orthopedic surgeries.
- Performed clinical validation in the operating room and evaluated the functionality and reliability of the device.

Rehabilitation Center, Second People's Hospital — Shenzhen, China
Student Assistant — *Aug 2022*

- Assisted doctors in rehabilitation evaluations and facility setup for patient rehabilitation.
- Acquired familiarity with basic rehabilitation facilities and procedures.

Awards & Achievements

International Symposium on Medical Robotics Best Student Paper Award 2026
Biomimetic Intelligence and Robotics Annual Best Paper Award 2024
Excellent Graduation Thesis Award (Top 10) Jul 2024
Outstanding Student Scholarship of Zhicheng College 2021, 2022, 2023
"Internet Plus" Innovation and Entrepreneurship Competition Aug 2023, Jul 2022
Excellent Team of "Sycamore Tree" Competition May 2022
RoboMaster 2021 Robotics Mecha Master Super Competition (First Prize) Sep 2021
"Fu Turui" Fellowship, Dept. of Biomedical Engineering (Top 3) Oct 2021
First Prize Scholarship for Freshmen (Top 4) Sep 2020

Skills & Knowledge

- **Laboratory:** FEM analysis, 3D printing, PCB design, Laser cutting, Circuit analysis, Motion capture.
- **Programming:** Python (Data analysis and PyTorch), C (Embedded system), MATLAB, Simulink, Rust, Java.
- **Software:** ANSYS, Opensim, SolidWorks, Arduino, EasyEDA/Altium, Unity, Blender.
- **Tools:** Git, Neovim, Docker, Jupyter, VS Code, Adobe PS, Final Cut Pro, Microsoft Office.
- **Languages:** Chinese (Native speaker), English (Proficient), Japanese and French (Elementary).

STEM & Community Outreach

Georgia Tech CEISMC Summer Program Leader Jul 2025
Guide and supervise K-12 students on campus, and introduce them to professional collage study.